HARSHINEE NADIMINTY

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Summary

Graduate student in Data Science, Analytics, and Engineering at Arizona State University with experience in backend development, machine learning, natural language processing, data analysis, and dashboard reporting. Skilled in building data-driven applications using Python, SQL, Django, Streamlit, Power BI, and modern ML frameworks, with hands-on work in NLP chatbots, RAG systems, recommendation systems, predictive modeling, and data visualization.

Education

M.S. in Data Science, Analytics & Engineering @ Arizona State University, Tempe **GPA: 3.5**
Aug 2025 – Present
B.Tech in Computer Engineering@ Pillai College of Engineering, University of Mumbai **CGPA: 8.6/10**

Experience

Nanobios Lab, IIT Bombay, India Dec 2023 – Apr 2024
Backend Developer Intern

- Developed backend systems using Django REST Framework to support research applications and improve platform reliability.
- Designed SQLite database structures, REST APIs, forms, and calculators to enable secure multi-user application functionality.
- Collaborated with cross-functional teams on backend architecture, documentation, ML-integrated features, and OpenAI-based components.

Projects

RAG (Retrieval-Augmented Generation) System for NLP Research Papers Aug 2025 – Dec 2025

- Built an end-to-end RAG pipeline over 1K+ arXiv NLP papers and 13K+ chunks for semantic search and citation-grounded QA.
- Evaluated SciBERT, MiniLM, MPNet, and Instructor embeddings, identifying SciBERT as strongest for domain-specific retrieval.
- Implemented FAISS + BM25 hybrid retrieval, improving result relevance with sub-6 ms query latency.

Smart Recipe Recommender Aug 2025 – Dec 2025

- Built a content-based recommendation system using TF-IDF and Sentence-BERT for personalized recipe suggestions.
- Designed an NLP pipeline with normalization, synonym mapping, lemmatization, to improve recommendation accuracy.
- Developed filtering logic for dietary restrictions and ingredient preferences to deliver relevant, health-conscious recommendations.

NLP Chatbot for National Education Policy in Hindi Jun 2024 – Nov 2024

- Developed a Hindi NLP chatbot for National Education Policy queries using preprocessing and vector-based retrieval.
- Implemented tokenization, normalization, LangChain, and ChromaDB for context-aware Hindi question answering.
- Built document-based retrieval workflows to deliver accurate, policy-focused responses in Hindi.

Radisson Blu Hotel Management System Jan 2024 - March 2024

- Created a hotel management system using Python, Tkinter, and MySQL to support room booking, meal booking, and data management.
- Designed user-friendly interfaces for customers and staff, automating booking workflows and reducing manual errors.
- Integrated a MySQL database to securely store user, booking, and operational information.

Publications

Sung-Geet: Emotion-Based Music Recommender JETIR, 2024

- Proposed a CNN-based emotion recognition system using MobileNet and MTCNN for real-time facial expression classification.
- Developed a Flask-based application to generate mood-based Hindi and English music recommendations.
- Integrated face detection, emotion classification, and recommendation logic into a unified recommendation pipeline.

Skills

- **Programming & Tools:** Python, SQL, Django, Django REST Framework, Streamlit, Flask, Tkinter, Git, Excel
- **Data & AI:** Machine Learning, Deep Learning, Natural Language Processing, Data Analysis, Data Preprocessing, Predictive Analytics, Recommendation Systems, Computer Vision
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, LangChain, ChromaDB, FAISS
- **Models & Techniques:** CNN, SVM, Logistic Regression, TF-IDF, Sentence-BERT, BM25, Vector Retrieval
- **Visualization & Databases:** Power BI, Tableau, DAX, MySQL, SQLite, SQL Server Management Studio